

The oligomeric state of chitooligosaccharide deacetylase from the marine bacterium *Vibrio campbellii*

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ABSTRACT The crystal structure of a chitooligosaccharide deacetylase (COD) from the marine bacterium *Vibrio campbellii* (formerly *V. harveyi*), referred to as *VhCOD*, exhibits a homodimeric assembly, consistent with previously reported X-ray structures of related *Vibrio* CODs. Here, we investigated whether this dimeric assembly is retained in solution. Native polyacrylamide gel electrophoresis (native PAGE) analysis revealed that *VhCOD* is monomeric in solution, with dimeric species present only as a minor population. Small-angle X-ray scattering further indicated that *VhCOD* preferentially adopts a monomeric state under dilute solution conditions. Analysis of the published crystal structures using PISA/jsPISA (Protein Interfaces, Surfaces and Assemblies/JavaScript version) predicted relatively weak monomer-monomer interactions and suggested that homodimer formation occurs only at sufficiently high protein concentrations ($>5 \text{ mg} \cdot \text{mL}^{-1}$). Chemical cross-linking experiments showed that macromolecular crowding reagents promote the formation of cross-linked dimer species. Taken together, we concluded that homodimerization of *Vibrio* CODs arises under crowded conditions rather than representing a dominant state. This interpretation is consistent with the physiological context of *VhCOD* as a secretory protein that is translocated through the crowded periplasm, where transient dimerization may be relevant prior to final export across the outer membrane.

WHY IT MATTERS Crystal structures of three *Vibrio* chitooligosaccharide deacetylases (CODs) show a similar homodimeric assembly. Similarity in the related homodimeric assemblies suggests a potential functional relevance. Here, native PAGE and small-angle X-ray scattering in solution revealed that CODs are predominantly monomeric under dilute conditions. PISA analysis and cross-linking experiments suggested that homodimerization is concentration dependent, occurring primarily at high protein concentrations or in crowded environments. COD dimerization appears to be promoted by macromolecular crowding, suggesting a physiological role as a secretory protein in the crowded periplasm, where transient dimerization may occur before export.

INTRODUCTION

Chitin deacetylases (CDAs; EC 3.5.1.41) and chitooligosaccharide deacetylases (CODs; EC 3.5.1.105) catalyze the removal of *N*-acetyl groups from chitin, a linear polysaccharide of *N*-acetylglucosamine (GlcNAc), and from its corresponding oligosaccharides, respectively, yielding acetate and the deacetylated sugar products.¹ These enzymes belong to the carbohydrate esterase family 4 (CE4) in the CAZy database: <http://www.cazy.org/CE4.html> and contain a conserved catalytic domain (CatD), known as the

NodB domain, composed of six to eight β strands surrounded by α helices. At the catalytic center, two histidine residues and one aspartate residue coordinate a Zn^{2+} cation.^{2,3} CDAs and CODs from different organisms exhibit distinct site specificities, acting on *N*-acetyl groups at preferred positions along the substrate and thereby generating diverse sugar sequences composed of GlcNAc and glucosamine (GlcN).^{1–4} To date, CDAs and CODs have been extensively investigated with respect to their modes of action, substrate specificities, and physiological roles, particularly in fungi.⁵

CODs from marine bacteria, including *Shewanella woodyi* (SwCOD),⁶ *Shewanella baltica* (SbCOD),⁷ *Vibrio cholera* (VcCOD),⁴ and *Vibrio parahaemolyticus* (VpCOD),⁸ have also been studied with respect to their

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structure and function. CODs from *Vibrio* species are secretory enzymes that produce β -D-N-acetylglucosaminyl-D-glucosamine (GlcNAc-GlcN), a key signaling molecule that enhances the chitin catabolic cascade.^{8,9} Furthermore, GlcNAc-GlcN has been shown to regulate flagellar rotation in *Vibrio* bacteria.¹⁰ Thus, CODs in *Vibrio* species are of physiological importance due to their ability to produce functional oligosaccharides.

X-ray crystallographic studies of VcCOD and VpCOD revealed that both enzymes consist of three domains: a CE4 CatD, followed by two carbohydrate-binding module family 12 (CBM12) domains.^{4,8} Despite being crystallized under different conditions and belonging to different space groups and unit cells (Table S1), the two enzymes adopt similar homodimeric assemblies in the crystal. The crystal structure of a COD from *V. campbellii* (formerly *V. harveyi*)-type strain BAA 1116 (VhCOD), which exhibits the same domain organization (Figures S1A and S1C), was also reported to form a homodimeric crystal structure (Figure S1B).¹¹ The amino acid sequence of VhCOD is highly similar to VcCOD and VpCOD, with sequence identities of 85.07% to VcCOD and 80.74% to VpCOD. VhCOD comprises a typical CE4 CatD complete with two histidyl and an aspartic side chain coordinated with the Zn^{2+} cation at the catalytic center (Figure S1D).

Homologous proteins can form conserved physiological homo-oligomers; however, crystal packing can also give rise to nonphysiological oligomeric assemblies that depend on crystallization conditions and space group. Although the three *Vibrio* CODs share highly similar amino acid sequences, they were crystallized under distinct conditions and in different space groups. It is therefore notable that all three structures exhibit related homodimeric assemblies, suggesting a potential functional relevance of dimer formation. This observation prompted us to investigate whether the homodimer observed in the crystal structures represents a stable assembly in solution and to elucidate the structural basis of homodimer formation, which may influence the enzymatic and biological functions of *Vibrio* CODs.¹²

Here, we produced recombinant VhCOD using an *Escherichia coli* expression system and analyzed its structure in solution by size-exclusion chromatography (SEC), native polyacrylamide gel electrophoresis (native PAGE), and small-angle X-ray scattering (SAXS). Protein-protein interface analysis was performed based on the crystal structures using PDBePISA/jsPISA.^{13,14} Furthermore, we conducted chemical cross-linking experiments using glutaraldehyde in the presence of crowding reagents. Based on these data, we discuss the molecular basis of homodimer formation observed in the crystal struc-

tures and its relevance under solution and crowded conditions.

MATERIALS AND METHODS

Materials

A synthetic *cod* gene encoding 427 amino acids of VhCOD was obtained from Gene Synthesis Service (Genscript, Hong Kong). Ni-NTA resin was a product of Genscript (Genscript), and Superdex-S200 Prep Grade resin for enzyme purification was purchased from GE Healthcare Life Sciences (Global Life Sciences Solutions, USA). Reagents used for the protein expression and purification were analytical grade and commercially available.

Protein expression and purification

Expression and purification of VhCOD were conducted based on the method reported previously.¹¹ Briefly, the full-length COD gene (*cod*) was cloned into the pQE60 expression plasmid (pQE60-*cod*), which was transformed into the *Escherichia coli* M15 (pREP) cells. Transformants were cultured in Luria Broth medium supplemented with $100\ \mu\text{g}\cdot\text{mL}^{-1}$ ampicillin and $25\ \mu\text{g}\cdot\text{mL}^{-1}$ kanamycin and grown at 37°C until the optical density at 600 nm reached 0.6–0.8. The *cod* gene expression was induced with $50\ \mu\text{M}$ isopropyl- β -galactoside (IPTG), and the culture was incubated for an additional 16 h at 18°C . Cells were harvested and resuspended in lysis buffer (20 mM Tris-HCl [pH 7.4], 100 mM NaCl, 5 units/mL DNase I, 0.1% [v/v] Triton X-100, and 1 mM phenylmethylsulphonyl fluoride [PMSF]) before lysis with an ultrasonicator. Cell debris was removed by centrifugation, and the supernatant was applied to Ni-NTA affinity chromatography. The partially purified protein was further purified by gel-filtration chromatography on a Superdex-200 column equilibrated with 20 mM Tris-HCl (pH 7.5) containing 100 mM NaCl. The purified VhCOD protein exhibited a single band at 45 kDa on SDS-PAGE.

Molecular weight determination

Size exclusion chromatography (SEC) was conducted using a Superdex-S200 column equilibrated with 20 mM Tris-HCl (pH 7.5) containing 100 mM NaCl at a flow rate of $0.4\ \text{mL}\cdot\text{mL}^{-1}$ using an ÄKTA Gold FPLC system. Four standard proteins, glucose oxidase (160 kDa), bovine serum albumin (65 kDa), ovalbumin (43 kDa), and RNase A (14 kDa), were used to estimate the molecular weight (MW) of VhCOD in solution. To calculate the distribution coefficient (K_{av}), blue dextran (1,000 kDa) was used to obtain the void volume (V_o), and N^ϵ -DNP-L-lysine hydrochloride

(348.7 Da) was used to obtain the bed volume of the stationary phase (V_t). Blue dextran and N^ε-DNP-L-lysine were mixed with *VhCOD* and applied to the column. The distribution coefficient (K_{av}) was calculated according to the following equation,^{15,16}

$$K_{av} = \frac{V_e - V_o}{V_t - V_o},$$

where V_e is the elution volume of each protein. The logarithmic molecular weight ($\log_{10}MW$) of the standard proteins were plotted against K_{av} to generate a calibration curve.

The molecular mass of *VhCOD* was also determined by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS). Sinapinic acid (12.5 mg·mL⁻¹) mixed with acetonitrile and 0.1% (v/v) trifluoroacetic acid (TFA) was used as the matrix. A 0.7 μL aliquot of the matrix was applied to the sample plate and mixed with 0.3 μL of 50 μM *VhCOD* before drying. Mass spectra were obtained in positive ion reflection mode over an m/z range 30–210 kDa using an Autoflex TOF/TOF (Bruker Co.).

Native polyacrylamide gel electrophoresis

Native polyacrylamide gel electrophoresis (Native PAGE) analyses were performed using 6% and 8% polyacrylamide gels. *VhCOD* samples at different concentrations (500, 100, 50, 10, 1, and 0.05 μM) were loaded on the gels. Electrophoresis was carried out at 100 V for 2 h 45 min at 4°C using running buffer containing 0.375 M Tris-HCl (pH 8.8), 30% (w/v) acrylamide, 10% (w/v) ammonium persulfate, and Tetramethylethylenediamine (TEMED). Gels were stained using a standard Coomassie brilliant blue protocol.

Small angle X-ray scattering

The *VhCOD* solutions (4.0, 5.7, and 10.3 mg·mL⁻¹) were exposed to X-ray radiation with a wavelength (λ) of 1.38 Å at 10°C for 10 min. Measurements of the empty cell and buffer were collected for background subtraction. One-dimensional scattering profiles, $I(q)$, were obtained by azimuthal averaging of two-dimensional detector images using SAXSIT (Synchrotron Light Research Institute, Nakhon Ratchasima, Thailand) with *BioXTAS RAW*.¹⁷ The momentum transfer q was calculated as $q = 4\pi \sin\theta/\lambda$, where 2θ is the scattering angle ranging from 0.02 to 0.39 Å⁻¹. Pair distribution functions $P(r)$ were calculated from the scattering curves using *GNOM* from the *ATSAS* package (v.3.1.3).¹⁸

Experimental scattering curves were compared with theoretical curves calculated using *CRY SOL*

from monomeric and dimeric models of *VhCOD*.¹⁹ *Ab initio* shape reconstruction was performed using *DAM-MIF*²⁰ with 20 independent bead models without imposed symmetry (P1), and the models were averaged using *DAMMAVER*. Superposition of the predicted monomeric and dimeric structures with the Small angle X-ray scattering (SAXS)-derived envelopes was performed using *SASpy*,²¹ and alignment quality was assessed using the normalized spatial discrepancy (NSD) value.

Interface analysis and detection of *VhCOD* homodimerization

To analyze the state of the interface between the two *VhCOD* molecules in the homodimeric structure, the *VhCOD* structures were analyzed using PDBePISA: <https://www.ebi.ac.uk/pdbe/pisa/>.¹³ The jsPISA server: <https://www.ccp4.ac.uk/pisa/>¹⁴ was used to predict the oligomeric state of *VhCOD* at different protein concentrations.

Cross-linking under crowding conditions

Cross-linking experiments were performed using PEG 3350, PEG 2000, and glycerol as molecular crowding agents. Glutaraldehyde was used as the cross-linking reagent at a final concentration of 2% (v/v) in reactions containing 2.5 mg·mL⁻¹ *VhCOD* in the 20 mM potassium phosphate (pH 7.4), with or without molecular crowding reagents. Reaction mixtures were incubated at 25°C for 30 min and terminated with 1× SDS-loading buffer. Samples were analyzed by SDS-PAGE. The intensity of protein bands corresponding to dimeric *VhCOD* was quantified using ImageJ.²²

RESULTS AND DISCUSSION

Purified *VhCOD* preparation

Histidine-tagged *VhCOD* was purified by sequential Ni-NTA affinity chromatography followed by gel-filtration chromatography and exhibited a single protein band at approximately 45 kDa on SDS-PAGE (Figure 1A, inset). MALDI-TOF-MS gave a single mass signal at $m/z = 45,654.364$ (Figure 1A), which is similar to the calculated molecular mass of 45,311 Da based on the amino acid sequence. The slightly higher molecular mass (by 343 Da) may be due to matrix adducts.

Purified *VhCOD* was applied to SEC on a Superdex-S200 column using blue dextran and N^ε-DNP-L-lysine as V_o and V_t references, respectively. Four MW standards were used to calibrate the column, and the chromatographic profiles were superimposed (Figure 1B). Based on the calibration curve obtained from plotting K_{av} values against $\log_{10} MW$, the apparent MW of

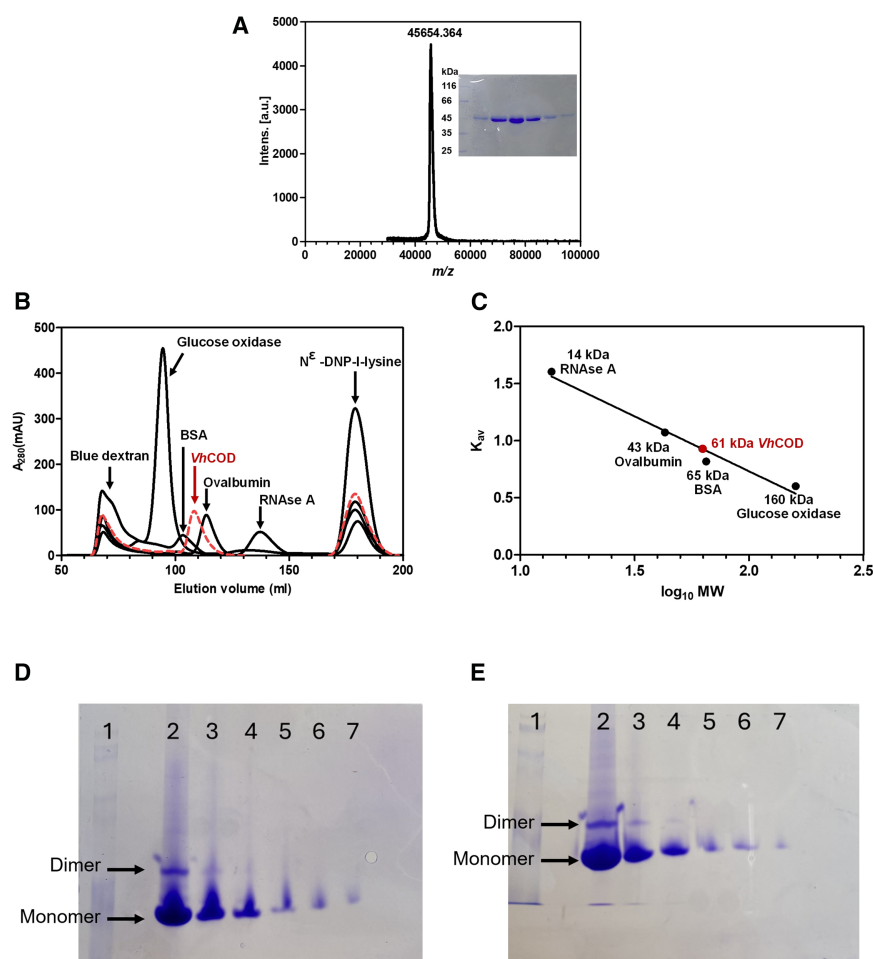


Figure 1. Molecular weight determination of purified *VhCOD*. (A) MALDI-TOF-MS spectrum of purified *VhCOD*. The mass signal observed at m/z 45,654.364 corresponds to the molecular mass calculated from the amino acid sequence. Inset, SDS-PAGE profile of the purified *VhCOD*. (B) SEC chromatogram of *VhCOD* (red dashed line) and MW standards on a Superdex-S200 Prep Grade column. Blue dextran and N^ε-DNP-L-lysine (348.7 Da) were used to determine V_t and V_o . (C) MW determination of *VhCOD* based on the calibration curve. (D and E) Native PAGE analysis of the dimeric/monomeric state of *VhCOD* with 6% and 8% acrylamide gel, respectively. Different concentrations of native *VhCOD* were loaded onto the gels. Lane 1, MW standards. Lanes 2–7, 500 (22.5 mg · mL⁻¹), 100 (4.5 mg · mL⁻¹), 50 (2.25 mg · mL⁻¹), 10 (0.45 mg · mL⁻¹), 1 (0.045 mg · mL⁻¹), and 0.05 (0.0025 mg · mL⁻¹) μM, respectively.

VhCOD was estimated to be 61 kDa (Figure 1C). This value is ambiguous with respect to the oligomeric state (Figure 1C) because the monomer and dimer are 45 and 90 kDa, respectively. Given that the molecular shape of *VhCOD* was not globular, the Stokes radius is unlikely to correlate directly with molecular mass, which may explain the overestimation of the monomer molecular weight. Alternatively, partial interactions between *VhCOD* and the chromatographic resin may contribute to the apparent retardation.

Then, we conducted native PAGE (Figures 1D and 1E). Increasing concentrations of purified *VhCOD* were loaded onto the polyacrylamide gel to observe the dimeric/monomeric state. The protein band corresponding to the dimeric state was observed in higher concentrations above 2.25 mg · mL⁻¹, whereas the monomeric form predominated at all concentrations tested.

Small angle X-ray scattering

To examine the state of *VhCOD* in solution, which reflects the biological environment,²³ SAXS analyses

were conducted on *VhCOD* solutions (SASBDB code: SASDZZ9) at three different concentrations (4.0, 5.7, and 10.3 mg/mL). The resulting scattering curves and their corresponding Kratky plots are shown in Figures S2A and S2B, respectively. The Kratky plots exhibited “bell-shaped” profiles in the lower qR_g region but lacked shoulder peaks in the higher qR_g region. This is characteristic of a fully folded, globular conformation for *VhCOD* under these conditions.

The analysis of the scattering curves by means of a Guinier plot revealed a linear relationship in each protein concentration, indicating the absence of aggregation or interference between the *VhCOD* molecules in each concentration (Figure 2A). The shape and size of the protein were determined from the radius of gyration (R_g) and maximal particle dimension (D_{max}), which were obtained from the slope of the Guinier plot and the pair distribution function $P(r)$, respectively (Figure 2B). The R_g values obtained were 27.2, 27.0, and 27.2 Å; the D_{max} values were 81, 79, and 82 Å for the individual *VhCOD* solutions (4.0, 5.7, and 10.3 mg · mL⁻¹), respectively. From the correlation volume, we calculated the MW as 43, 45, and

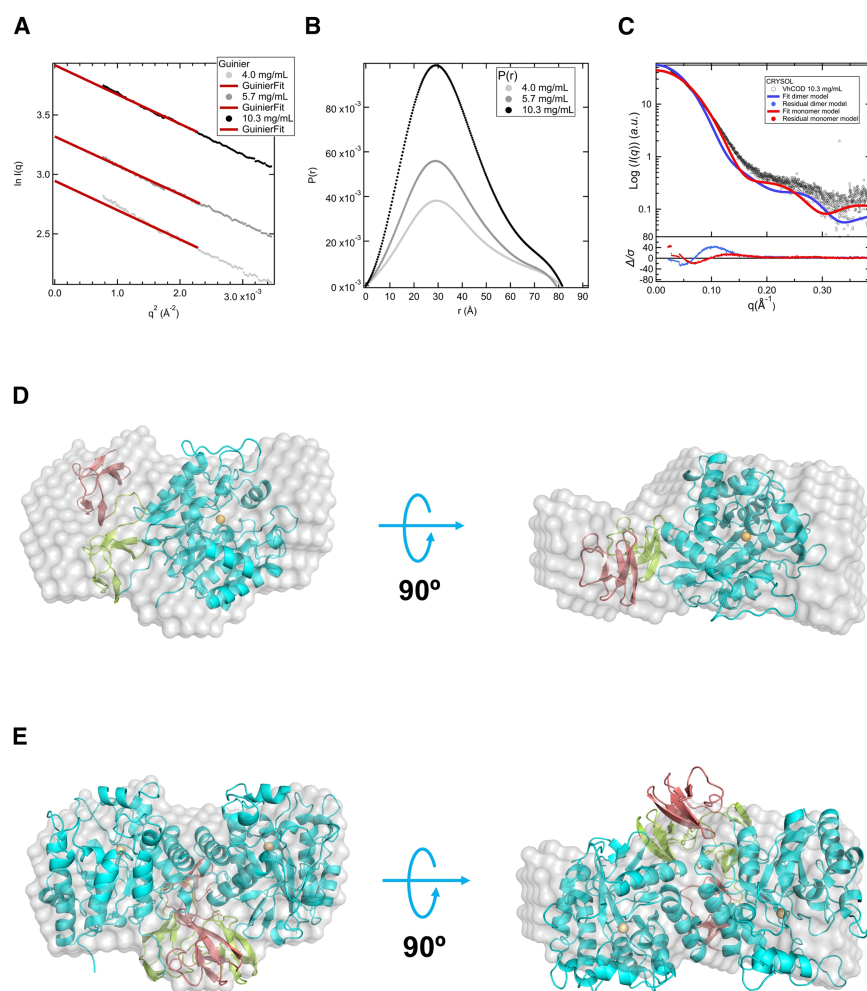


Figure 2. SAXS analysis of VhCOD (SASBDB code: SASDZZ9). (A) Guinier plot at concentrations of 4.0, 5.7, and 10.3 mg · mL⁻¹. The straight red lines represent Guinier fits. (B) The pair distribution function $P(r)$. Experimental data points are indicated by gray dots as in (A). (C) Top: *CRYSOLE* analysis of the scattering curve (10.3 mg · mL⁻¹) with both the monomeric (red) and homodimeric (blue) models. Bottom: error weight difference between the theoretical and experimental data (monomer model, red dots; dimer model, blue dots). (D) *Ab initio* molecular envelope obtained from *DAMMIF/DAMMAVER* superimposed with the crystal structure of the VhCOD monomer (PDB: 8YFP). (E) *Ab initio* molecular envelope superimposed with the VhCOD dimer (PDB: 8YFP).

48 kDa.²⁴ These values suggested a compact globular structure for VhCOD,²⁵ supporting the finding from the Kratky plot (Figure S2B). The calculated molecular mass of VhCOD (43–48 kDa) was consistent with the mass spectrum (Figure 1A) and the calculated mass of the monomeric VhCOD based on the amino acid sequence (45 kDa).

The X-ray scattering curve obtained at the highest enzyme concentration (10.3 mg · mL⁻¹) was further analyzed using the program *CRYSOLE*, and the results are shown in Figure 2C. The quality of the fits for the monomeric and dimeric models was assessed using an error-weighted residual difference plot of $\Delta/\sigma = [I_{\text{exp}}(q) - I_{\text{mod}}(q)]/\sigma(q)$ versus q (bottom, Figure 2C). The monomeric model (red) showed a relatively even distribution around zero, indicating a superior fit.²⁶ χ^2 for the monomer model was 34.5, while that for the dimer model was 57.6, supporting the monomeric state in solution. Furthermore, *ab initio* modeling was conducted using *DAMMIF* and *DAMMAVER*, using 20 independent bead models without imposed symmetry (*P1*). The superimposition of the

crystal structure of monomeric VhCOD (generated from an Ensemble Optimization Method [EOM] of the VhCOD structure, PDB: 8YFP) onto the SAXS-derived averaged/filtered molecular envelope demonstrated a better fit of the crystal structure with the envelope derived from the SAXS data collected in the monomeric state than in the dimeric state (Figure 2D). For the VhCOD monomer, 20 independent *DAMMIF* runs showed high convergence with a mean NSD of 0.734 (SD = 0.090) and a stable χ^2 range of 1.59–1.73. All 20 models were grouped into a single cluster, confirming a unique and reliable solution. The resolution of the resulting shape, calculated via SASRES, is 31 ± 3 Å.²⁷ The data listed in Table S2 strongly suggest that VhCOD predominantly exists in a monomeric form in solution.

Interface analysis

To evaluate the strength and structural characteristics of the homodimer interface observed in the crystal structures, we analyzed protein-protein interactions

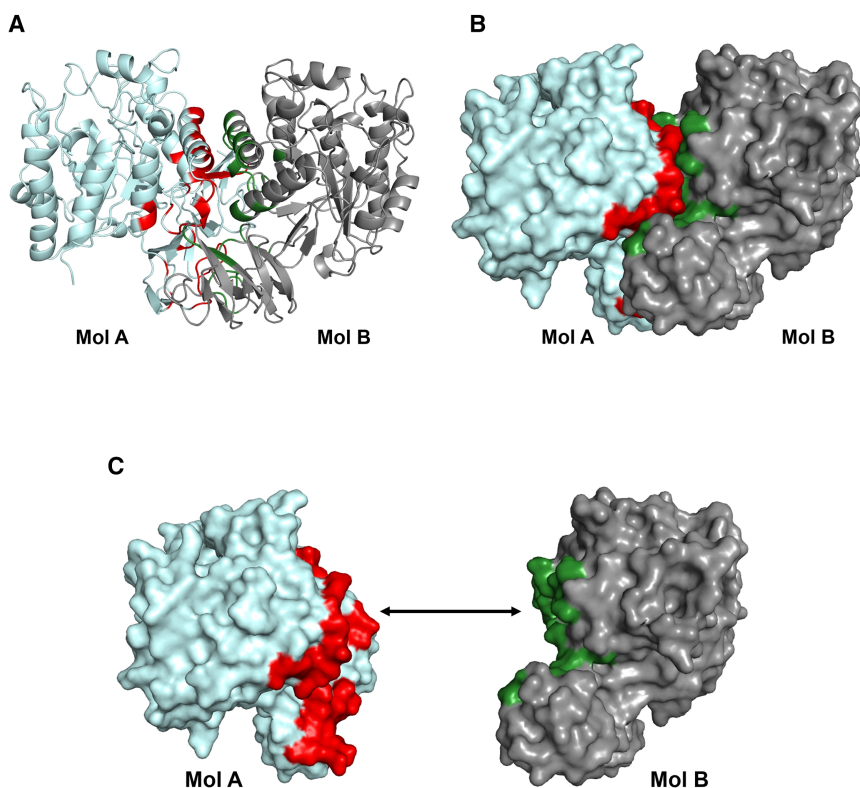


Figure 3. Interface analysis of the homodimeric *VhCOD*. (A and B) Cartoon (A) and surface (B) illustrations of the interaction surface between Mol A (red) and Mol B (green) of *VhCOD*. (C) The interaction surface when they are dissociated.

using computational and experimental approaches.¹³ The crystal structures of *VhCOD* (PDB: 8YFP), *VcCOD* (PDB: 4NY2), and *VpCOD* (PDB: 3WX7) were analyzed using the PDBePISA server. This analysis provided estimates of buried surface area, accessible surface area, and solvation free energy changes upon dimerization, enabling comparison among the three homologous enzymes (Table S3).

The complexation significance scores (CSS), which reflect the contribution of the interface to overall binding stability,²⁸ were low for all three enzymes: 0.182 for *VhCOD*, 0.172 for *VcCOD*, and 0.146 for *VpCOD*, indicating weak intermolecular interactions. Structural analysis of *VhCOD* showed that both the CatD and CBM domains were found to participate in dimerization. In CatD, 22 residues are involved in the interactions between the monomers. In CBM12-1, 15 residues participate in interface formation, whereas CBM12-2 contributes with only two residues. Overall, the interface area of each monomer accounts for only 8.2%–8.3% of the total surface area (red and green regions of Figure 3; Table S3). Similar interface characteristics were observed for *VcCOD* and *VpCOD* (Table S3). Nevertheless, the PISA-calculated free energy of dissociation (ΔG_{diss}) is 12.8 kcal·mol⁻¹, suggesting a tight interaction between the monomers. In this context, it would be important to experimentally determine ΔG_{diss} . We roughly assessed the K_d from the

native PAGE profile as follows: from Lane 2 of Figure 1D (500 μM), the dimer intensity was much lower than that of the monomer. When we postulated, based on the PAGE profile, that the intensity ratio is 0.5/9.5 (dimer/monomer) at 500 μM , the K_d was estimated to be 1.3 μM , corresponding to approximately 8 kcal/mol, suggesting a weak interaction. This experimental value is below the PISA-derived ΔG_{diss} (12.8 kcal/mol), consistent with known limitations of PISA-based estimations: https://www.ebi.ac.uk/pdbe/pisa/pi_disclaimer.html. Thus, the native PAGE data clearly indicate that *Vibrio* CODs form homodimers through weak intermolecular interactions. These CODs appear to preferentially adopt a monomeric state in solution.

Concentration dependence of the dimer formation of *VhCOD*

The oligomeric state of *VhCOD* as a function of protein concentration was further predicted using jsPISA.¹⁴ As shown in Figure S3, the analysis suggested that the homodimer becomes favored at protein concentrations more than the crossing point, 10^{-6} – 10^{-5} M. The crystallization was conducted at 5 mg·mL⁻¹ (approximately 10^{-4} M), which is higher than that of the crossing point. To experimentally test whether crowded conditions enhance dimer formation, we performed chemical cross-linking experiments using

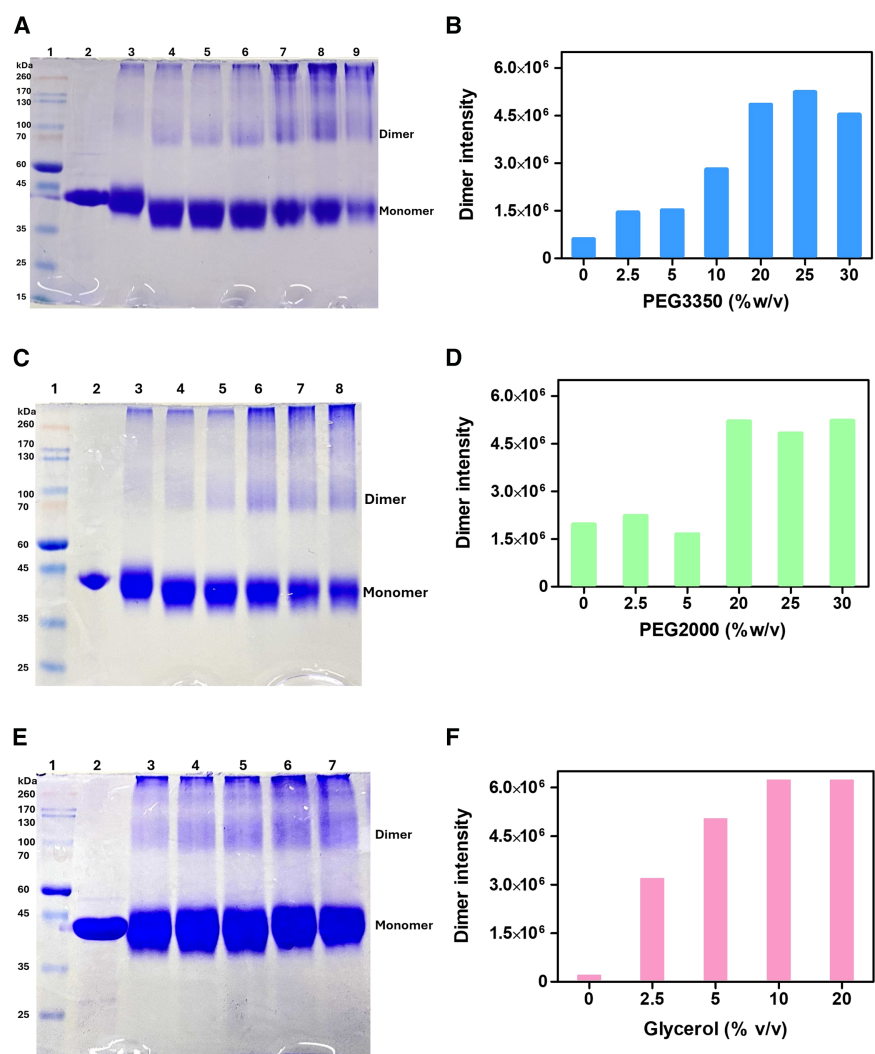


Figure 4. Cross-linking analysis of VhCOD. The experiments were performed using glutaraldehyde as a cross-linker in the presence of the crowding reagents, PEG 3350, PEG 2000, and glycerol. (A, C, and E) SDS-PAGE profiles of VhCOD cross-linked in the presence of PEG 3350, PEG 2000, and glycerol, respectively. Lane 1, protein ladder; lane 2, VhCOD without glutaraldehyde and crowding reagent; lanes 3–9, VhCOD (2.5 mg·mL⁻¹), 2% glutaraldehyde, and increasing concentrations of the crowding reagents. (B, D, and F) Effect of crowding reagent concentrations on the dimer band intensities, which were quantified by ImageJ based on the SDS-PAGE profiles shown in (A), (C), and (E).

glutaraldehyde in the presence of three crowding reagents. The presence of PEG 3350, PEG 2000, or glycerol enhanced dimer formation at a VhCOD concentration of 2.5 mg·mL⁻¹ (Figure 4). Increasing concentrations of crowding reagents further promoted dimerization, indicating that spatial confinement facilitates protein-protein interactions. These results suggest that dimer formation is favored not only under crystallization conditions but also in crowded environments such as the periplasm.

CONCLUSION

VhCOD is a secretory enzyme synthesized in the cytoplasm and translocated through the periplasm, where it encounters a densely crowded environment.²⁹ All experimental evidence obtained from native PAGE, SAXS, PDBePISA/jsPISA analysis, and cross-linking experiments (Figures 1, 2, 3, and 4) consistently indi-

cates that VhCOD preferentially adopts a monomeric form rather than the dimer in dilute solution due to the weak interaction between the monomers. Homodimer formation is observed only under highly crowded conditions, such as those present in protein crystals or mimicked by macromolecular crowding reagents. Moreover, the catalytic center is located far from the dimerization interface, making it unlikely that dimer formation directly affects catalytic efficiency. Therefore, any impact of dimerization on the catalytic function is expected to be minimal. Consequently, biochemical data obtained for VhCOD in solution, including enzyme kinetics and ligand binding, should be interpreted as the monomeric form to accurately reflect the enzyme function. However, VhCOD's role is as a secretory protein, during which dimerization may be relevant as it translocates through the crowded periplasm before final export across the outer membrane.

DATA AVAILABILITY

The authors have deposited the SAXS data in the SASBDB database. SASBDB accession code: SASDZZ9 (<https://www.sasbdb.org/data/SASDZZ9/7o47ln1mai>)

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AUTHOR CONTRIBUTIONS

Conceptualization, W.S.; methodology, R.C.R. and N.K.; investigation, S.P., R.C.R., and N.K.; data curation and analysis, S.P., T.F., R.C.R., N.K., and W.S.; writing – original draft, S.P., R.C.R., T.F., and W.S.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.bpr.2026.100275>.

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Supplemental information

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Supporting materials

The oligomeric state of chitooligosaccharide deacetylase from the marine bacterium *Vibrio campbellii*

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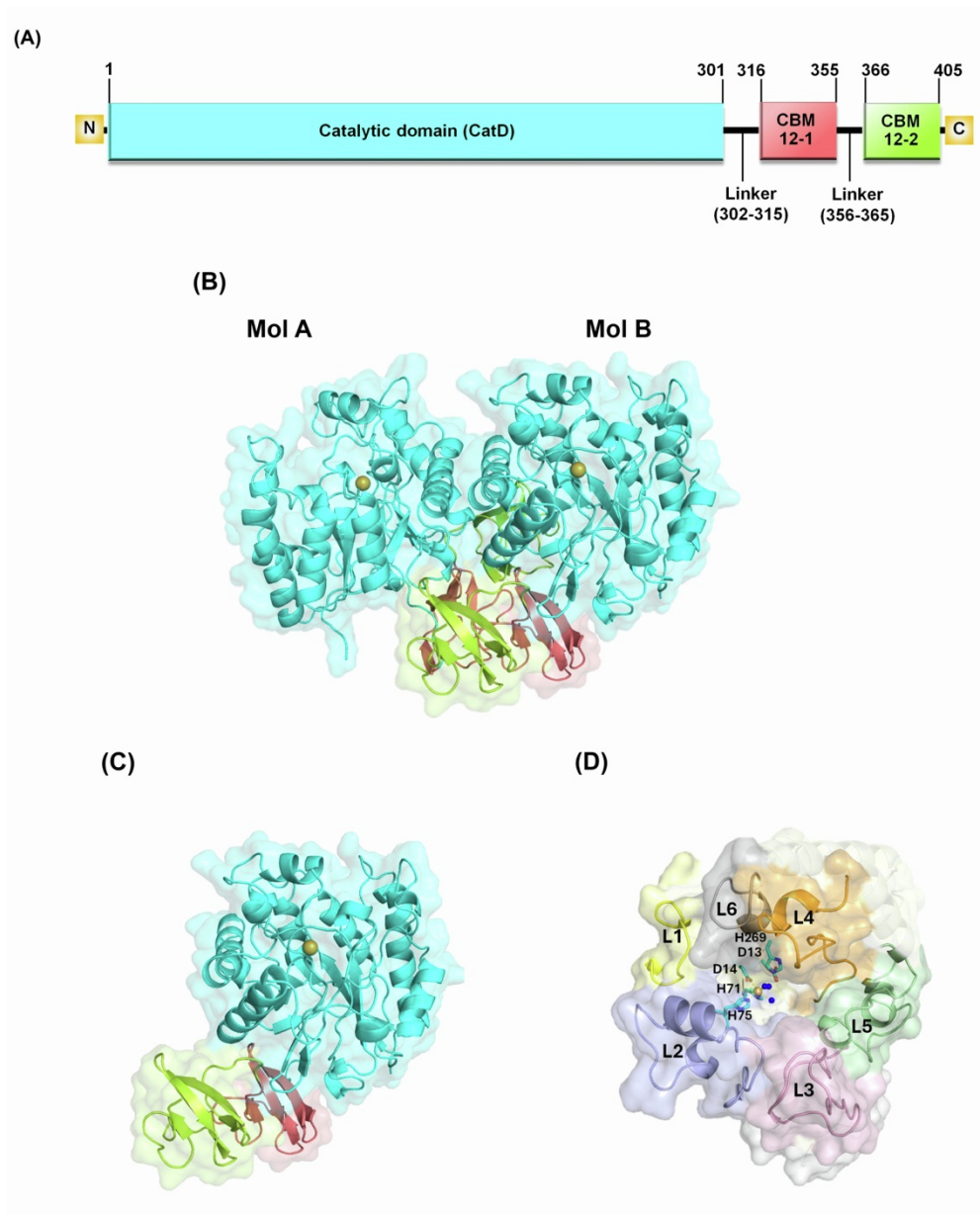
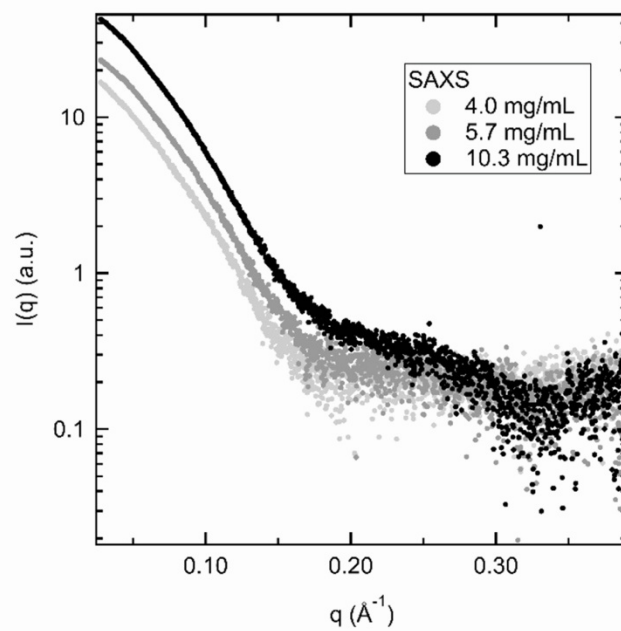


Fig. S1. (A) Schematic representation of the domain organization of *VhCOD*. The CE4 catalytic domain (CatD, residues 1-301) and two carbohydrate binding modules family12 domains, CBM12-1 and CBM12-2 (residues 316-355 and 366-405), are shown in cyan, pink and green, respectively. (B) Homodimeric structure of *VhCOD*. (C) Monomeric structure of *VhCOD*. (D) Cartoon and surface illustrations of the six loops in CatD. Individual loops (L1-L6) are colored in yellow, purple, pink, orange, light green, and gray, respectively. In the catalytic core surrounded by L1-L6 (six loop structures), a zinc ion (gold sphere) is coordinated with Asp14, His71, and His75. Asp13 and His269 act as the catalytic base and acid, respectively.

(A)



(B)

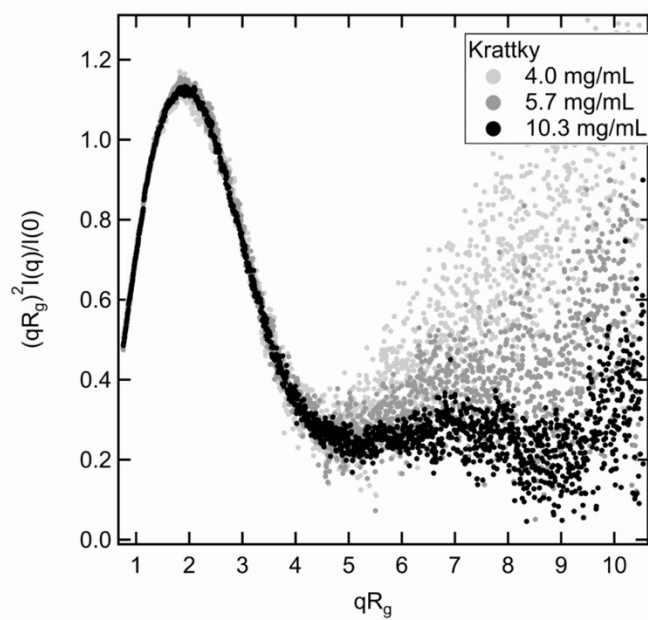


Fig. S2. SAXS analysis of VhCOD in three concentrations (4.0, 5.7, and 10.3 mg·mL⁻¹) (A) Scattering profiles (B) Kratky analysis.

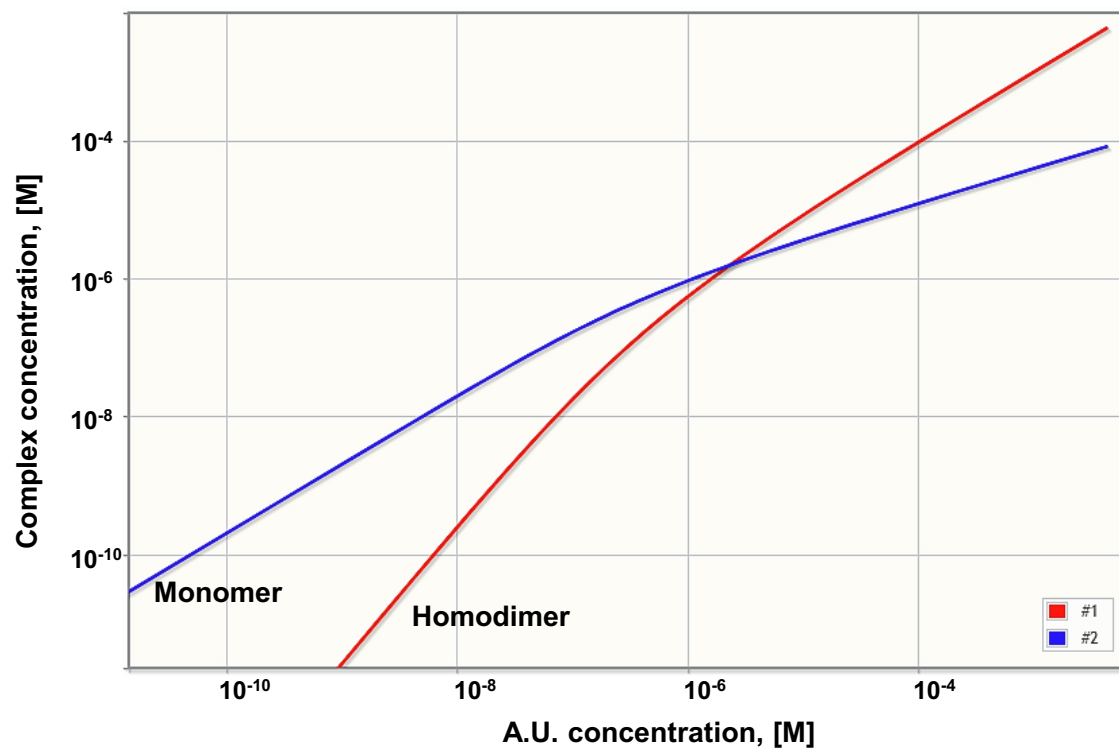


Fig. S3. jsPISA prediction of the oligomeric state of VhCOD (PDB ID :8YFP). Red line: homodimer (#1), and blue line: monomer (#2).

Table S1. Crystal structure details of *Vh*COD compared with *Vc*COD and *Vp*COD obtained from PDB database

PDB ID	8YFP	4NY2	3WX7
Organism	<i>Vibrio harveyi</i>	<i>Vibrio cholerae</i>	<i>Vibrio parahaemolyticus</i>
Crystal data			
Space group	I 2 ₁ 3	P 1 2 ₁ 1	P 2 ₁ 2 ₁ 2 ₁
Number of molecules in asymmetric unit	2	2	2
Unit cell			
Length (Å)	(a, b, c) = 209.061, 209.061, 209.061	(a, b, c) = 61.9, 112.54, 69.36	(a, b, c) = 61.819, 104.106, 140.378
Angle (°)	(α, β, γ) = 90	(α, β, γ) = 90, 94.05, 90	(α, β, γ) = 90
Crystallization experiment			
pH	7.5	6.5	7.0
Temperature (K)	292	291	298
Condition			
Buffer system	0.1M buffer system2 (1M Sodium HEPES, MOPS (acid) pH 7.5)	80mM MES pH 6.5	0.1M Tris-HCl pH 7.0
Salt system	0.06M Divalent (0.3M Magnesium chloride hexahydrate, 0.3M Calcium chloride)	0.16M Ca (OAc) ₂	0.2M MgCl ₂
Precipitant system	35% precipitant mix 4 (25% v/v MPD, 25% PEG 100, 25% w/v PEG 3350)	14.4% (w/v) PEG 8000 and 20% glycerol	10% PEG8000

Table S2. SAXS data adhering to the IUCrJ guidelines (28)

Parameter	VhCOD (Monomer)
DAMMIF (default parameters, 20 calculations)	
q range for fitting ($\text{\AA}^{-1}$)	0.0296–0.2544
Symmetry, anisotropy assumptions	P1, none
NSD (standard deviation), No. of clusters	0.734 (0.090), 1
χ^2 range	1.59-1.73
Resolution (from SASRES) (Tuukkanen et al., 2016) ($\text{\AA}$)	31 ± 3
Exposure time / Number of frames	1 x 600s
Radiation damage assessment	Negligible; confirmed via R_g independence across multiple concentrations and prior beamline validation (29)

Table S3. Assembly and interface information of the *Vh*COD homodimer compared with *Vc*COD and *Vp*COD obtained from PDBePISA analysis.

PDB ID	8YFP (<i>Vh</i> COD)		4NY2 (<i>Vc</i> COD)		3WX7 (<i>Vp</i> COD)	
CSS	0.182		0.172		0.146	
Multimeric state	2		2		2	
Copies in unit cell	24		2		4	
Formula	$A_2a_2b_2c_3$		$A_2a_2b_2c_2d_4$		$A_2a_2b_2c_2d_4$	
Composition	AB[Zn] ₂ [MG] ₂ [CA] ₃		AB[Zn] ₂ [CA] ₂ [ACT] ₂ [GOL] ₂		AB[CA] ₂ [Zn] ₂ [ACT] ₂ [GOL] ₄	
Dissociation pattern	A[Zn][MG] ₂ [CA] + B[Zn][CA]		A[CA][Zn][ACT][GOL] + B[CA][Zn][ACT][GOL]		A[CA][Zn][ACT][GOL] ₂ + B[CA][Zn][ACT][GOL] ₂	
Surface area, Å ²	30975.3		30699.1		30431.5	
Buried area, Å ²	3474.0		4671.5		4655	
ΔG_{int} , kcal·mol ⁻¹	-130.0		-116.1		-124	
ΔG_{diss} , kcal·mol ⁻¹	12.8		8.8		7.3	
$T\Delta S_{diss}$, kcal·mol ⁻¹	14.3		14.4		14.3	
Symmetry number	2.0		2.0		2.0	
Interface	MolA	MolB	MolA	MolB	MolA	MolB
Number of atoms						
interface	149 (4.8)	154 (4.9)	170 (5.4)	164 (5.2)	152 (4.8)	150 (4.8)
surface	1631 (52)	1615 (52)	1668 (53)	1643 (52)	1611 (51)	1621 (52)
total	3131 (100)	3125 (100)	3170 (100)	3160 (100)	3135 (100)	3135 (100)
Number of residues						
interface	41 (10)	42 (10)	46 (11)	45 (11)	42 (10)	44 (11)
CatD	21	22	25	24	20	20
CBM12-1	14	15	18	18	17	18
CBM12-2	6	5	5	5	5	5
surface	356 (88)	356 (89)	361 (89)	368 (91)	357 (88)	356 (89)
total	403 (100)	402 (100)	405 (100)	405 (100)	402 (100)	402 (100)
Solvent accessible area, Å²						
interface	1391.3 (8.2)	1407.2 (8.3)	1548.2 (9.0)	1545.6 (9.1)	1347.6 (8.0)	1325.2 (7.9)
total	16925.8 (100)	16859.4 (100)	172252 (100)	16976.5 (100)	16775.1 (100)	16709 (100)

The values of parentheses are percentages of the individual parts.